THE HONG KONG POLYTECHNIC UNIVERSITY
香港理工大學

Undergraduate Research and Innovation Scheme (URIS)
Research Project Proposal Form

[Up to four A4 pages (single-line spaced)]

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1. **Project Title:**

Data-Driven Dynamic State Estimation for Situational Awareness of Modern Power Grids

2. Abstract:

The transition to smart grids introduces variability and uncertainty in power system operations, driven by the integration of distributed energy resources (DERs), electric vehicles (EVs), and solar panels. This presents hurdles for grid operators maintaining reliable electricity supply. Traditional static state estimation methods, particularly Weighted Least Squares (WLS), face increasing challenges due to their reliance on accurate physical models and inability to predict future grid conditions. This project aims to design a data-driven dynamic state estimation (DSE) framework that leverages Deep Learning to enable predictive situational awareness. Unlike static WLS, the proposed method learns the temporal mapping between historical measurement sequences and future bus voltages directly from simulated operational data. The principal objectives are to: (1) implement a deep learning model (such as DNN/LSTM) for multi-step voltage forecasting; (2) evaluate model robustness against measurement loss and bad data injection under dynamic load conditions; and (3) extend the analysis to predictive situational awareness by visualizing future voltage deviations and flagging potential violations ahead, enabling proactive grid monitoring.

3. Project Keywords:

Power Systems, Dynamic State Estimation, Deep Learning, Predictive Situational Awareness, Low Observability

4. Project Background:

The transition to smart grids is accelerating in Hong Kong, driven by rapid adoption of electric vehicles (EVs), rooftop solar panels, and other emerging applications. As of late 2025, Hong Kong had approximately 149,000 EVs on the road, with over 70% of new private car registrations being electric [1]. Solar adoption is also growing quickly, with major installations now generating enough electricity to power over 1,200 households annually [2]. This growth creates challenges for the distribution grid. Unlike centralized power plants, these technologies introduce stochastic, bidirectional power flows and voltage fluctuations. Grid operators need visibility to prevent overloads and maintain reliability.

Traditional state estimation, primarily Weighted Least Squares (WLS), has been the standard approach in Energy Management Systems for decades. WLS uses physical grid models to calculate voltages and power flows from sensor measurements [3]. However, research shows WLS struggles in modern distribution grids because it requires many well-placed sensors and accurate network models: conditions rarely met in practice [4]. When measurements are lost or network configuration changes unexpectedly, WLS often fails to produce reliable estimates.

Deep Neural Networks (DNNs) can overcome these limitations by learning relationships between available measurements and grid conditions directly from data [5]. Unlike WLS, a trained DNN or LSTM can estimate the grid's state even when sensors are offline or topology changes. This "soft sensing" approach uses AI to estimate unmeasured quantities cost-effectively [6].

However, static state estimation provides only a snapshot of the present moment. In modern grids, operators require foresight. Dynamic State Estimation (DSE) addresses this by incorporating temporal correlations in measurement data. By processing past measurements, a DSE model can forecast near-future voltage profiles, giving operators critical lead time to mitigate issues before violations occur. This project leverages the inherent memory of recurrent neural networks (RNNs) to perform DSE, moving beyond static interpolation to predictive situational awareness.

Table 1: Comparison of WLS and DNN-Based State Estimation Approaches

Criterion	Weighted Least Squares (WLS)	Deep Neural Network (DNN)
Model Requirement	Requires accurate network parameters and topology model	Learns directly from historical or simulated data
Observability	Fails when measurement set is insufficient	Can interpolate in low-observability scenarios
Bad Data Handling	Post-estimation residual tests; vulnerable to gross errors	Potential robustness when trained with noisy or corrupted data scenarios
Computational Speed	Iterative Newton-Raphson (multiple steps)	Single forward pass after offline training
Topology Changes	Requires explicit model updates	Adapts if represented in training data

Recent studies have demonstrated deep learning's potential for state estimation. Azimian et al. (2022) showed that DNNs can estimate states in unobservable distribution systems [7], while several projects have demonstrated the ability of LSTM networks in effectively capturing spatio-temporal dynamics for forecasting both present and future distribution system states [8] [9]. However, these studies primarily focus on estimation accuracy as an end. This project builds upon this foundation in two ways. First, it explicitly applies dynamic DNN-based estimation to EV-driven load variability in distribution grids, forecasting voltage trends rather than merely reconstructing present states. Second, it synthesizes state estimation with predictive situational awareness visualization, moving beyond numerical accuracy to actionable operational intelligence.

This project aligns with Research Focus 4 of PolyU's Research Centre for Grid Modernisation (RCGM): "AI for Future Smart Grid," in which Prof. Bu is actively involved [10].

5. Project Objectives:

Objective 1: To develop and train a Deep Learning model for Dynamic State Estimation (DSE) using time-series simulated operational data from a representative distribution feeder (IEEE 33-bus system). The model will learn to forecast future voltage profiles from historical measurement sequences, establishing baseline forecasting accuracy against a persistence model and a static DNN benchmark.

Objective 2: To evaluate the robustness of the proposed DSE model under challenging dynamic grid scenarios, specifically measurement loss and bad data injection occurring during time-varying load ramps (e.g., EV charging peaks). The analysis will quantify the model's resilience in low-observability conditions where traditional static WLS fails and where forecasting errors could lead to undetected violations.

Objective 3: To extend the dynamic state estimation framework to predictive situational awareness by developing a visualization module that interprets forecasted voltage trajectories, identifies impending constraint violations, and issues early warnings (e.g., undervoltage expected

in 5 minutes during EV peak). This demonstrates practical applicability for proactive real-world grid monitoring.

6. Project Design and Method:

To ensure the proposed framework can perform reliably, the DSE model must be trained and tested on realistic, time-varying distribution grid scenarios. The model must learn to map a sequence of past sparse measurements (e.g., the last 5 timestamps) to the future voltage profile across all buses. Therefore, it is essential that the model can maintain accurate forecasts even when measurements are intermittently lost or corrupted during critical ramping events.

Simulation Framework and Data Generation: A simulation framework using Pandapower and Python will be implemented. The IEEE 33-bus radial distribution feeder will serve as the test network [11]. Time-series load profiles will be generated by combining base residential curves with synthetic EV charging load shapes derived from Hong Kong's EV adoption data. One hundred continuous operational trajectories will be simulated using time-series power flow, each producing a sequence of 144 ten-minute intervals. For each timestep, a full AC power flow calculation provides ground truth voltages. A realistic measurement configuration will be simulated, with sensors placed only at the substation and a few select nodes, mimicking sparse instrumentation.

Dynamic State Estimation Model: The core of the DSE framework is a sequence-to-sequence recurrent neural network, implemented using TensorFlow/PyTorch. This architecture is grounded in the principle that recurrent networks (LSTM/GRU) excel at capturing temporal dependencies in time-series data, enabling the model to learn how past measurements evolve into future states.

- **Input:** A sliding window of T_past measurement vectors.
- **Architecture:** Two stacked LSTM layers (64 and 32 units) followed by fully connected layers with ReLU (Rectified Linear Unit) activation.
- **Output:** Voltage magnitudes (and optionally phase angles) at all buses for T_future steps ahead.
- **Training:** Mean Squared Error (MSE) loss and Adam optimizer, with 80/20 train-test split on the 100 daily trajectories.

Benchmark Comparisons: The DSE model's forecasting performance will be compared against:

- **Persistence Model:** Assumes voltage remains constant at the last known value.
- **Static DNN:** A feed-forward network that predicts the next step from a single snapshot.
- **WLS (Static):** Demonstrates that even with perfect static estimation, future states cannot be known without a forecasting engine.

Predictive Situational Awareness Visualization: A module developed using Matplotlib/Plotly will map the forecasted voltage profiles onto the network diagram. It will implement a predictive alert system. This provides operators with an actionable lead time to make corrective measures.

7. Research Activities:

Project Period	Tasks and Deliverables
1st – 12th month	**Sep 1, 2025 – Oct 31, 2025 (2 months):** Set up Python environment with Pandapower, TensorFlow/PyTorch. Model IEEE 33-bus system and implement time-series load profile generator with EV charging profiles.

	Nov 1, 2025 – Apr 30, 2026 (6 months): Generate full dataset of operational trajectories. Extract ground truth voltage sequences and simulate realistic sparse measurement streams. **May 1, 2026 – Aug 31, 2026 (4 months):** Implement baseline models. Design LSTM-based DSE architecture, define sliding window data preparation pipeline, and train initial forecasting models. Tune hyperparameters (window size, forecast horizon, layer sizes).
13th – 24th month	**Sep 1, 2026 – Dec 31, 2026 (4 months):** Evaluate DSE model robustness. Quantify forecasting accuracy metrics and compare lead-time advantage over baselines. **Jan 1, 2027 – Aug 31, 2027 (8 months):** Develop predictive situational awareness visualization module. Implement color-coded alert system. Demonstrate end-to-end workflow on peak EV charging scenarios.

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